

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
3 Maart 2016

Semester Test 1

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Module EBN111
Electricity and Electronics
3 March 2016



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: Exam information:

Maksimum punte: Maximum marks:	46	Volpunte: Full marks:	45
Duur van vraestel: Duration of paper:	90 minute 90 minutes	Oopboek / toeboek: Open / closed book:	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (including this page):		16 + Answer sheet	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): Lecturer(s):	1. G Hancke
	2. D. Bhatt
	3. D. le Roux

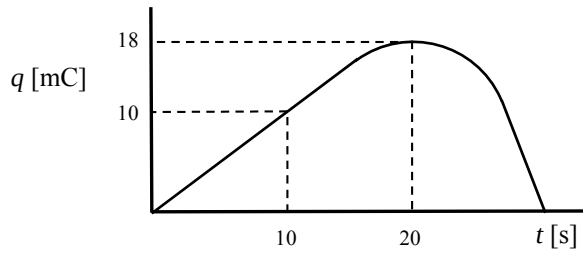
VRAAG/QUESTION 1 [12]

Vraag 1.1 [2]

Die lading wat in 'n geleier vloei word onder getoon. Bepaal die stroom by: a) $t = 10$ s en b) $t = 20$ s. (Die kurwe bereik sy maksimumwaarde by $t = 20$ s.)

Question 1.1 [2]

The charge flowing in a conductor is shown below. Determine the current at: a) $t = 10$ s en b) $t = 20$ s. (The curve reaches its maximum value at $t = 20$ s.)



Vraag 1.2 [2]

Die spanning oor 'n $5\ \Omega$ weerstand neem liniêr toe van 0 mV tot 15 mV tussen $t = 0$ s en $t = 20$ s. Bereken

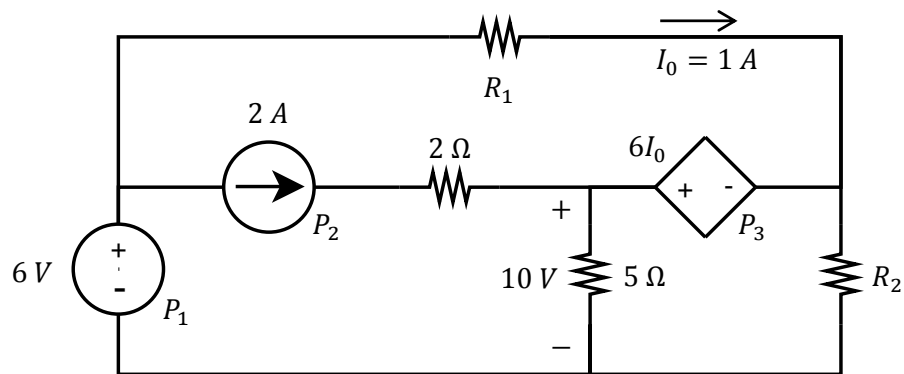
- Die waarde van die lading wat deur die weerstand vloei tussen $t = 5$ s en $t = 15$ s.
- Die drywing in die weerstand geabsorbeer by $t = 12$ s.

Question 1.2 [2]

The voltage across a $5\ \Omega$ resistor increases linearly from 0 mV to 15 mV between $t = 0$ s and $t = 20$ s. Calculate

- The value of the charge flowing through the resistor between $t = 5$ s and $t = 15$ s.
- The power absorbed by the resistor at $t = 12$ s.

Vraag 1.3 [3] 'n Elektriese verwarmer is gespesifiseer as 240 V; 3 kW. a) Hoeveel stroom trek die verwarmer? b) Hoeveel lading vloei deur dit in 1 minuut? c) Elektrisiteit kos R3/kWh. Bereken die totale koste oor een maand (30 dae) indien die verwarmer elke oggend vir 80 minute aangeskakel word?	Question 1.3 [3] An electric heater is specified as 240 V; 3 kW. a) What is the current flowing through the heater? b) How much charge flows through the heater in 1 minute? c) Electricity cost R3/kWh. Calculate the total cost if the heater is switched on for 80 minutes every morning for 1 month (30 days).
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Vraag 1.4 [5]Bepaal P_1 , P_2 , P_3 , R_1 en R_2 **Question 1.4 [5]**Determine P_1 , P_2 , P_3 , R_1 en R_2 .

VRAAG/QUESTION 2 [18]

Vraag 2.1 [4]

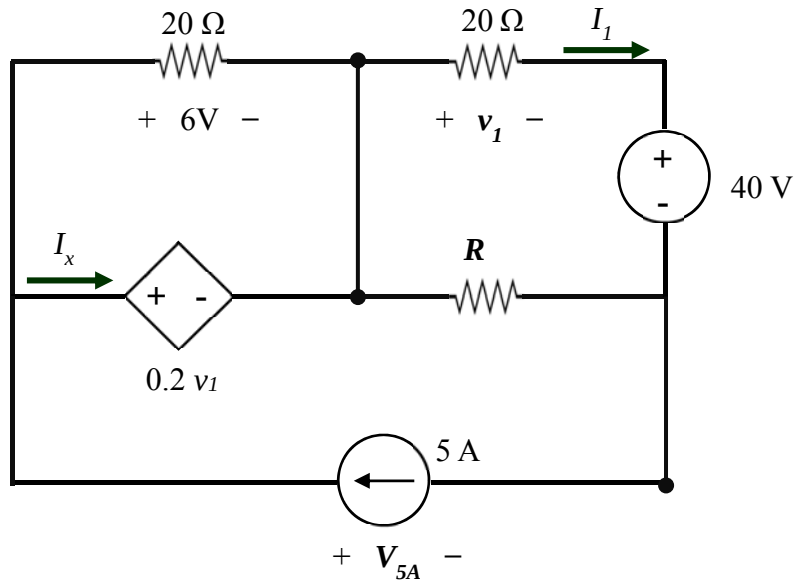
Vir onderstaande kringbaan, bereken die volgende:

- die stroom I_I ,
- die stroom I_x ,
- die spanning oor die 5 A stroombron, V_{5A} , en
- die waarde van die resistor R wat sal veroorsaak dat 'n stroom van 3.5 A daardeur vloei.

Question 2.1 [4]

Consider the circuit diagram shown below and calculate the following:

- the current I_I ,
- the current I_x ,
- the voltage across 5 A current source, V_{5A} , and
- the value of resistor R , that will allow a current of 3.5 A to flow through it.



Vraag 2.2 [4]

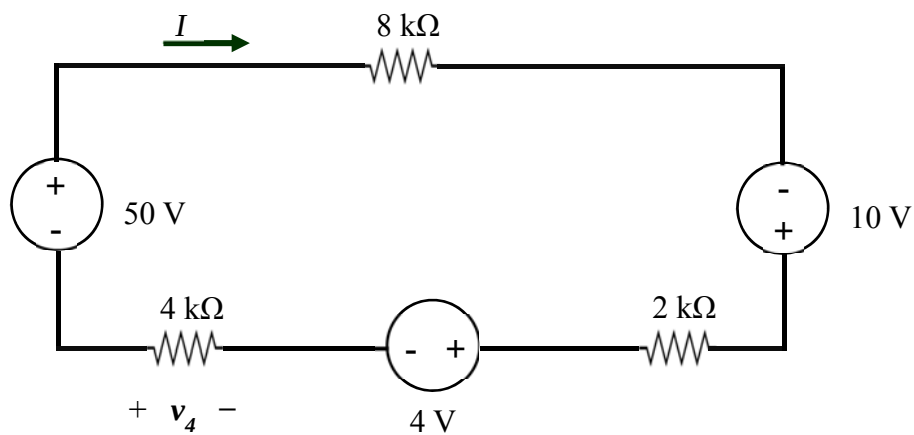
Bepaal die:

- a) spanning oor $4\text{ k}\Omega$ resistor, v_4 .
- b) stroom I .

Question 2.2 [4]

Determine the:

- a) voltage across $4\text{ k}\Omega$ resistor, v_4 .
- b) current I .



Vraag 2.3 [4]

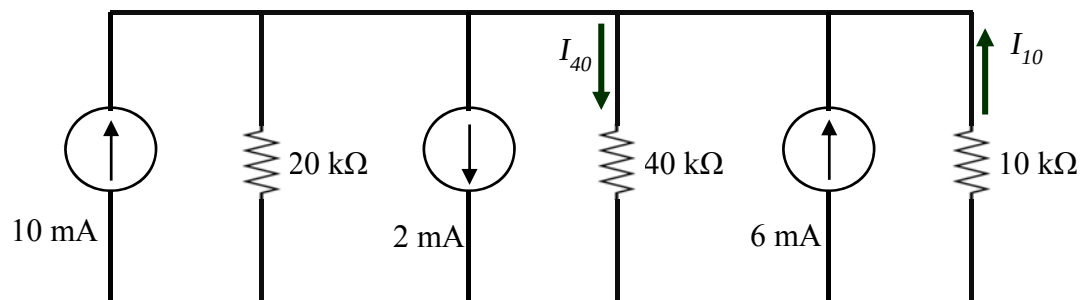
Bepaal die stroom deur

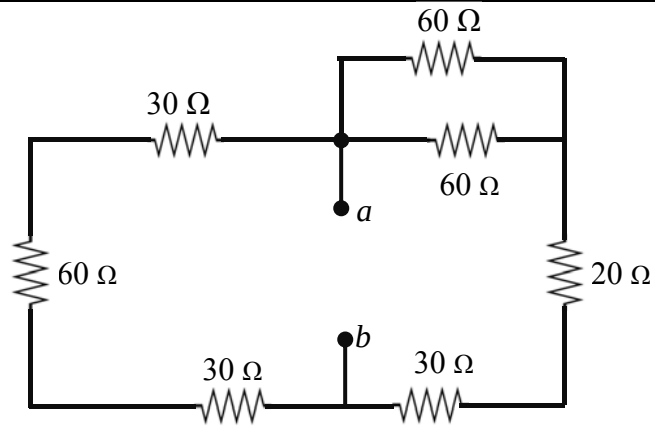
- a) die 40 k Ω resistor, I_{40} and
- b) die 10 k Ω resistor, I_{10} .

Question 2.3 [4]

Determine the current flowing through

- a) the 40 k Ω resistor, I_{40} and
- b) the 10 k Ω resistor, I_{10} .



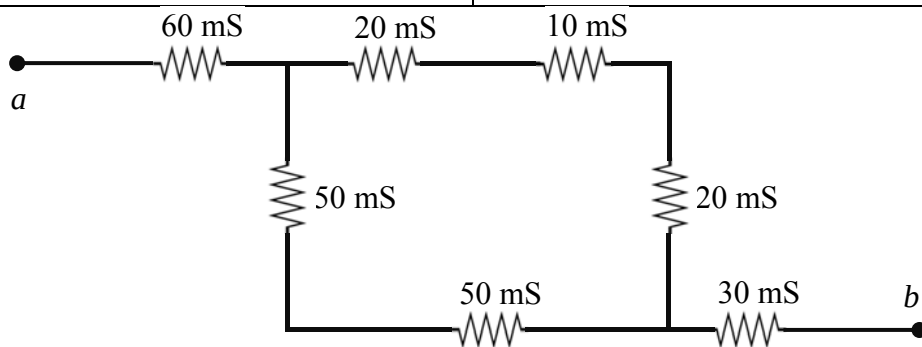
Vraag 2.4 [2]Bepaal die ekwivalente weerstand R_{ab} .**Question 2.4 [2]**Calculate the equivalent resistance R_{ab} .

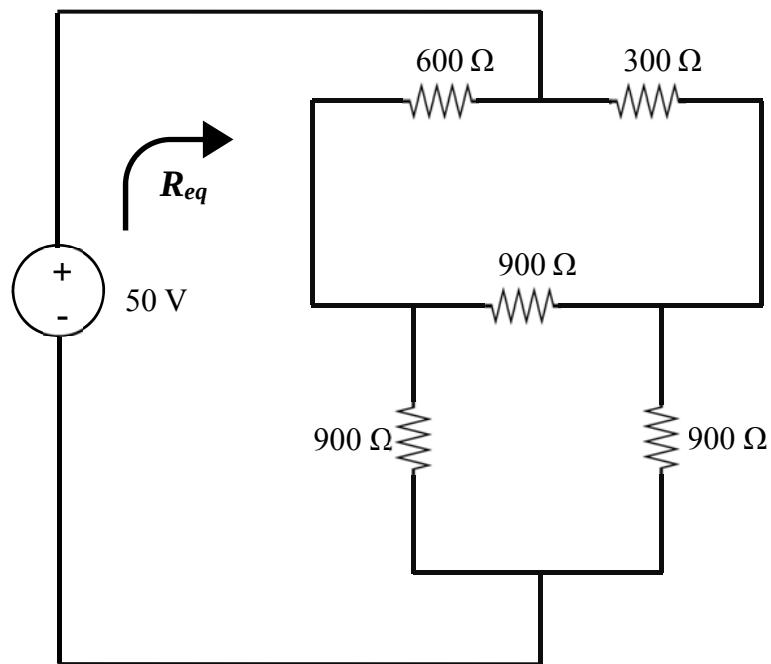
Vraag 2.5 [2]

Bepaal die ekwivalente geleiding, G_{ab} , tussen die nodes a - b vir onderstaande kringbaan.

Question 2.5 [2]

Calculate the equivalent conductance, G_{ab} , across the nodes a - b of the circuit below.



Vraag 2.6 [2]Bepaal die ekwivalente weerstand R_{eq} .**Question 2.6 [2]**Calculate the equivalent resistance, R_{eq} .

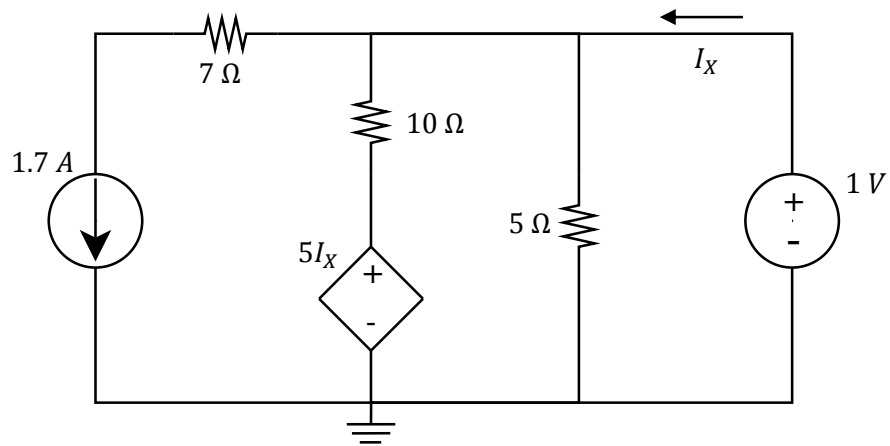
VRAAG/QUESTION 3 [16]

Vraag 3.1 [2]

Gebruik of node of lusanalyse en bepaal I_X .

Question 3.1 [2]

Use either nodal or mesh analysis to determine I_X .

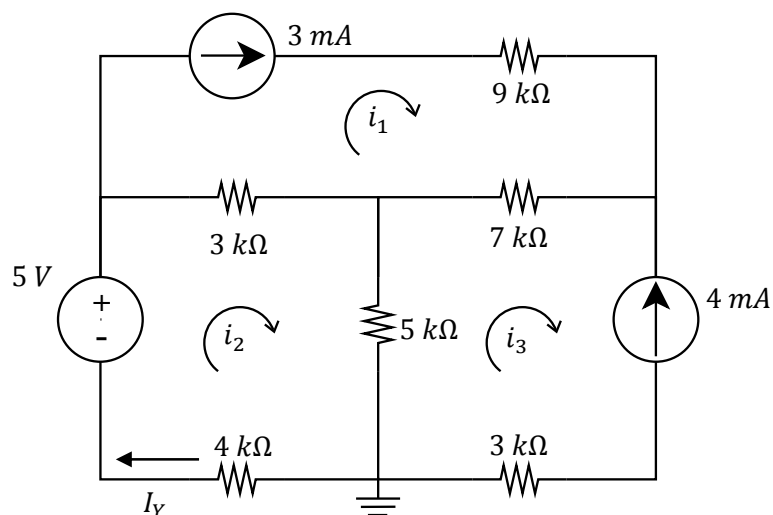


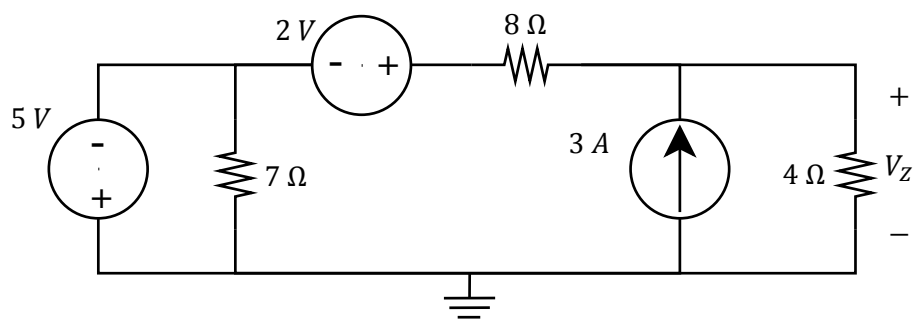
Vraag 3.2 [2]

Gebruik lusanalyse en bepaal I_Y .
(Wenk: $i_1 = 3 \text{ mA}$, $i_2 = I_Y$, en $i_3 = -4 \text{ mA}$.)

Question 3.2 [2]

Use mesh analysis to determine I_Y .
(Hint: $i_1 = 3 \text{ mA}$, $i_2 = I_Y$, and $i_3 = -4 \text{ mA}$.)



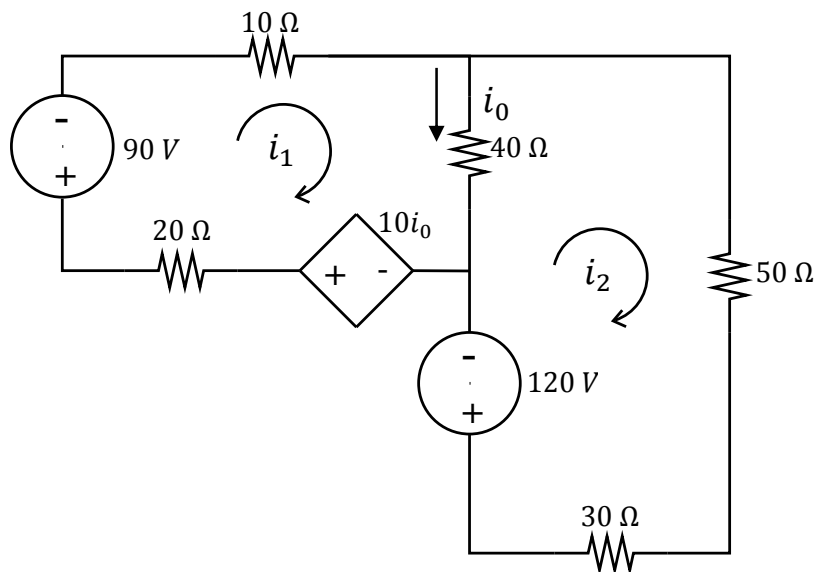
Vraag 3.3 [2]Gebruik node analyse en bepaal V_Z .**Question 3.3 [2]**Use nodal analysis to determine V_Z .

Vraag 3.4 [2]

Indien gegee dat $i_2 = -1.8 \text{ A}$, bepaal i_0 deur gebruik te maak van lusanalise.

Question 3.4 [2]

If given that $i_2 = -1.8 \text{ A}$, determine i_0 using mesh analysis.



Vraag 3.5 [4]a) Bepaal α indien $I_1 = 0.75$ A.

$$\begin{bmatrix} 4 & \alpha \\ -8 & 2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \end{bmatrix}$$

b) Bepaal β indien $v_1 = -3$ V en $v_3 = -1.8$ V.

$$\begin{bmatrix} 6 & 3 & -10 \\ -3 & 0 & \beta \\ 2 & -8 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$$

Question 3.5 [4]a) Determine α if $I_1 = 0.75$ A.

$$\begin{bmatrix} 4 & \alpha \\ -8 & 2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \end{bmatrix}$$

b) Determine β if $v_1 = -3$ V en $v_3 = -1.8$ V.

$$\begin{bmatrix} 6 & 3 & -10 \\ -3 & 0 & \beta \\ 2 & -8 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$$

Vraag 3.6 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$aV_1 + bV_2 = 30$$
$$cV_1 + dV_2 = -7.5$$

Question 3.6 [4]

Use nodal analysis to set up two equations of the following form:

$$aV_1 + bV_2 = 30$$
$$cV_1 + dV_2 = -7.5$$

